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(following lecture by A.P. Yauho)

Tunneling through a gap (Landauer formula)

L gap R
(no central region present!)

• Hamiltonian:

$$\begin{aligned} H &= H_C + H_T \\ H_C &= H_R + H_L = \sum_r (\epsilon_r - \mu) c_r^\dagger c_r + \sum_e (\epsilon_e - \mu) c_e^\dagger c_e \\ H_T &= \sum_{re} (t_{er} c_e^\dagger c_r + t_{re}^* c_r^\dagger c_e) \end{aligned} \quad (1)$$

The leads are decoupled, i.e.

$$\{c_e, c_{e'}^\dagger\} = \delta_{ee'}$$

$$\{c_r, c_{r'}^\dagger\} = \delta_{rr'}$$

$$\{c_r, c_e\} = \{c_r^\dagger, c_e^\dagger\} = \{c_r^\dagger, c_e\} = \{c_r, c_e^\dagger\} = 0$$

The latter follow from the general result for non-orthogonal orbitals:

$$\{c_i, c_j^\dagger\} = S_{ij}^{-1}$$

For decoupled sets L and R the block of $(S^{-1})_{LR} = 0$.

• The current in the left lead ($\hbar = 1$):

$$\hat{I}_L = -e \dot{N}_L = -ie [H, N_L] = -ie ([H_L, N_L] + [H_T, N_L])$$

Here: $N_L = \sum_e c_e^\dagger c_e$ and

$$[H_L, N_L] = 0$$

$$[H_T, N_L] = \sum_{re} \left(t_{er} [c_e^\dagger c_r, c_e^\dagger c_e] + t_{re}^* [c_r^\dagger c_e, c_e^\dagger c_e] \right)$$

$$\begin{aligned} \textcircled{1} &= c_e^\dagger c_r c_e^\dagger c_e - c_e^\dagger c_e c_e^\dagger c_r = c_e^\dagger c_e^\dagger c_e c_r - c_e^\dagger (\delta_{ee} - c_e^\dagger c_e) c_r \\ &= -c_e^\dagger c_e c_e c_r - \delta_{ee} c_e^\dagger c_r + c_e^\dagger c_e^\dagger c_e c_r = -\delta_{ee} c_e^\dagger c_r \end{aligned}$$

$$\begin{aligned} \textcircled{2} &= c_r^\dagger c_e c_e^\dagger c_e - c_e^\dagger c_e c_r^\dagger c_e = c_r^\dagger (\delta_{ee} - c_e^\dagger c_e) c_e - c_r^\dagger c_e^\dagger c_e c_e \\ &= \delta_{ee} c_r^\dagger c_e - c_r^\dagger c_e^\dagger c_e c_e - c_r^\dagger c_e^\dagger c_e c_e = \delta_{ee} c_r^\dagger c_e \end{aligned}$$

not everywhere
 ϵ_e or ϵ_r are
replaced by
 $\epsilon_e - \mu$ or $\epsilon_r - \mu$

$$\hookrightarrow [H_T, N_L] = \sum_{re} (t_{er} c_e^\dagger c_r + t_{re}^* c_r^\dagger c_e)$$

and hence

$$\hat{J}_L = -ie \sum_{re} (t_{re}^* c_r^\dagger c_e - t_{er} c_e^\dagger c_r)$$

$$\boxed{\hat{J}_L = -ie \sum_{re} (t_{re}^* \langle c_r^\dagger c_e \rangle - t_{er} \langle c_e^\dagger c_r \rangle)} \quad (2)$$

The averages are w.r.t exact density matrix $\hat{\rho}_t$:

$$\langle \hat{O} \rangle = \text{Tr}[\hat{\rho}_t \hat{O}] = \text{Tr}[\hat{\rho}_0 e^{iHt} \hat{O} e^{-iHt}] = \text{Tr}[\hat{\rho}_0 \hat{O}_H(t)],$$

where $\hat{\rho}_0$ is $\hat{\rho}_t$ at $t=0$, and $\hat{O}_H(t)$ is $\hat{O}$ in the Heisenberg representation with the full Hamiltonian.

We introduce the NEGF

$$G_{ij}^<(t, t') = -i \frac{\text{Tr}[e^{-\beta H} \hat{T}_c c_{H_i}(t) c_{H_j}^\dagger(t')]}{\text{Tr}[e^{-\beta H}]} = -i \langle \hat{T}_c c_{H_i}(t) c_{H_j}^\dagger(t') \rangle,$$

Since H does not depend on time, $G_{ij}^<(t, t') \equiv G_{ij}^<(t-t')$, depends only on the time difference. We only need

$$G_{ij}^<(t, t') = i \langle c_{H_j}^\dagger(t') c_{H_i}(t) \rangle \equiv -G_{ji}^<(t', t)^*$$

at equal times:

$$\langle c_r^\dagger c_e \rangle = -i G_{er}^<(t, t), \quad \langle c_e^\dagger c_r \rangle = -i G_{re}^<(t, t)$$

$$\begin{aligned} \hookrightarrow \hat{J}_L &= -ie \sum_{re} (t_{re}^* (-i) G_{er}^<(t, t) - t_{er} (-i) G_{re}^<(t, t)) = \\ &= -e \sum_{re} [-t_{re}^* G_{re}^<(t, t)^* - t_{er} G_{re}^<(t, t)] \end{aligned}$$

or

$$\boxed{\hat{J}_L = \frac{2e}{\hbar} \text{Re} \sum_{re} t_{er} G_{re}^<(t, t)} \quad (3)$$

We now need to calculate $G_{re}^<(t, t') \equiv G_{re}^<(t-t')$. We start from the EOM for $\langle \hat{T}_c c_{H_i}(t) c_{H_j}^\dagger(t') \rangle$

$$G_{ij}(t, t') = -i \langle \hat{T}_C c_{Hi}(t) c_{Hj}^{\dagger}(t') \rangle$$

$$i\partial_{t'} c_{Hj}^{\dagger}(t') = i\partial_{t'} e^{iHt'} c_j^{\dagger} e^{-iHt'} = ie^{iHt'} (iH c_j^{\dagger} - i c_j^{\dagger} H) e^{-iHt'}$$

$$= -e^{iHt'} [H, c_j^{\dagger}] e^{-iHt'}$$

$$[H, c_j^{\dagger}] i\partial_{t'} c_{Hj}(t') = -e^{iHt'} [H, c_j] e^{-iHt'}$$

For

$$G_{re}(t, t') = \theta(t-t') G_{re}^>(t, t') + \theta(t'-t) G_{re}^<(t, t')$$

$$= i\theta(t-t') \langle c_{Hr}(t) c_{He}^{\dagger}(t') \rangle + i\theta(t'-t) \langle c_{He}^{\dagger}(t') c_{Hr}(t) \rangle$$

we have to calculate,

$$[H, c_e^{\dagger}] = [H_L, c_e^{\dagger}] + [H_T, c_e^{\dagger}] =$$

$$[H, c_r] = [H_R, c_r] + [H_T, c_r] \quad (\leftarrow \text{not needed})$$

Here:

$$[H_L, c_e^{\dagger}] = \sum_{e'} \varepsilon_{e'} (c_{e'}^{\dagger} c_{e'} c_e^{\dagger} - c_e^{\dagger} c_{e'}^{\dagger} c_{e'}) =$$

$$= \sum_{e'} \varepsilon_{e'} [c_{e'}^{\dagger} (\delta_{ee'} - c_e^{\dagger} c_{e'}) + c_e^{\dagger} c_{e'}^{\dagger} c_{e'}] = \varepsilon_e c_e^{\dagger}$$

$$[H_T, c_e^{\dagger}] = \sum_{e', r'} [t_{e'r'} (c_{e'}^{\dagger} c_{r'} c_e^{\dagger} - c_e^{\dagger} c_{e'}^{\dagger} c_{r'}) + t_{r'e}^* (c_{r'}^{\dagger} c_{e'} c_e^{\dagger} - c_e^{\dagger} c_{r'}^{\dagger} c_{e'})]$$

$$= \sum_{e', r'} [t_{e'r'} (c_{e'}^{\dagger} c_{r'} c_e^{\dagger} + c_{e'}^{\dagger} c_e^{\dagger} c_{r'}) + t_{r'e}^* (c_{r'}^{\dagger} (\delta_{ee'} - c_e^{\dagger} c_{e'}) + c_e^{\dagger} c_{r'}^{\dagger} c_{e'})] = \sum_{r'} t_{re}^* c_r^{\dagger} \quad (4)$$

$$[H, c_e^{\dagger}] = \varepsilon_e c_e^{\dagger} + \sum_r t_{re}^* c_r^{\dagger}, \quad \boxed{i\partial_{t'} c_{He}^{\dagger}(t') = -\varepsilon_e c_{He}^{\dagger}(t') + \sum_r t_{re}^* c_{Hr}^{\dagger}(t')}$$

Next,

$$[H_R, c_r] = \sum_{r'} \varepsilon_{r'} (c_{r'}^{\dagger} c_{r'} c_r - c_r c_{r'}^{\dagger} c_{r'}) = \boxed{(\varepsilon_r c_r)}$$

$$= \sum_{r'} \varepsilon_{r'} (-c_{r'}^{\dagger} c_r c_{r'} - (\delta_{rr'} - c_{r'}^{\dagger} c_{r'}) c_{r'}) = -\varepsilon_r c_r$$

$$\begin{aligned}
 [H_T, c_r] &= \sum_{e'} \left[t_{e'r'} (c_{e'}^\dagger c_{r'} c_r - c_r c_{e'}^\dagger c_{r'}) + t_{r'e'}^* (c_{r'}^\dagger c_{e'} c_r - c_r c_{r'}^\dagger c_{e'}) \right] \\
 &= \sum_{e'} \left\{ t_{e'r'} (-c_{e'}^\dagger c_r c_{r'} + c_{e'}^\dagger c_r c_{r'}) + t_{r'e'}^* (-c_{r'}^\dagger c_r c_{e'} - (\delta_{r'e'} - c_{r'}^\dagger c_r) c_{e'}) \right\} \\
 &= - \sum_{e'} t_{r'e'}^* c_{e'}
 \end{aligned}$$

or $[H, c_r] = -\epsilon_r c_r - \sum_e t_{re}^* c_e$, $\boxed{i\partial_{t'} c_{Hr}(t') = \epsilon_r c_{Hr}(t') + \sum_e t_{re}^* c_{He}(t')}$

We thus have:

$$\begin{aligned}
 +i\partial_{t'} G_{re}^>(t, t') &= i(\delta(t-t') G_{re}^> + \delta(t'-t) G_{re}^<) \\
 &+ \theta(t-t')(i\partial_{t'} G_{re}^>) + \theta(t'-t)(i\partial_{t'} G_{re}^<) \\
 &= -\delta(t-t')(G_{re}^> - G_{re}^<) + \theta(t-t')(i\partial_{t'} G_{re}^>) + \theta(t'-t)(i\partial_{t'} G_{re}^<),
 \end{aligned}$$

with $G_{re}^> - G_{re}^< = -i \langle c_{Hr}^\dagger(t) c_{He}^\dagger(t') \rangle - i \langle c_{He}^\dagger(t') c_{Hr}(t) \rangle$

$$= -i \langle \{c_{Hr}(t), c_{He}^\dagger(t')\} \rangle = -i \langle \delta(t-t') \rangle_{re} = \delta(t-t') \delta_{re}$$

$\hookrightarrow i\partial_{t'} G_{re}^<(t, t') = \delta(t-t') + \theta(t-t')(i\partial_{t'} G_{re}^>) + \theta(t'-t)(i\partial_{t'} G_{re}^<)$

or $i\partial_{t'} G_{re}^<(t, t') = \delta(t-t') + \theta(t-t') \left(\frac{c_{Hr}^\dagger(t) c_{He}^\dagger(t')}{c_{He}^\dagger(t') c_{Hr}(t)} \right) (i\partial_{t'} G_{re}^>) + \theta(t'-t)(i\partial_{t'} G_{re}^<)$

where:

$$\begin{aligned}
 i\partial_{t'} G_{re}^>(t, t') &= -i \langle c_{Hr}(t) (i\partial_{t'} c_{He}^\dagger(t')) \rangle \\
 &= -i \langle c_{Hr}(t) (-\epsilon_e c_{He}^\dagger(t') - \sum_{r'} t_{re}^* c_{Hr'}^\dagger(t')) \rangle \\
 &= -\epsilon_e (-i) \langle c_{Hr}(t) c_{He}^\dagger(t') \rangle - \sum_{r'} t_{re}^* (-i) \langle c_{Hr}(t) c_{Hr'}^\dagger(t') \rangle \\
 &= \epsilon_e G_{re}^>(t, t') - \sum_{r'} t_{re}^* G_{rr'}^>(t, t')
 \end{aligned}$$

and

$$i\partial_{t'} G_{re}^<(t, t') = i \langle (i\partial_{t'} c_{He}^\dagger(t')) c_{Hr}(t) \rangle =$$

$$= i \left\langle \left(-\varepsilon_e c_{He}^+(t) - \sum_{r'} t_{re}^* c_{Hr'}^+(t') \right) c_{Hr}(t) \right\rangle$$

$$= -\varepsilon_e (+i) \langle c_{He}^+(t) c_{Hr}(t) \rangle - \sum_{r'} t_{re}^* (+i) \langle c_{Hr'}^+(t') c_{Hr}(t) \rangle$$

$$= -\varepsilon_e G_{re}^<(t, t') - \sum_{r'} t_{re}^* G_{rr'}^<(t, t').$$

$$\hookrightarrow -i \partial_{t'} G_{re}^<(t, t') = \delta(t-t') + \theta(t-t') \left[\varepsilon_e G_{re}^> + \sum_{r'} t_{re}^* G_{rr'}^> \right]$$

$$+ \theta(t'-t) \left[\varepsilon_e G_{re}^< + \sum_{r'} t_{re}^* G_{rr'}^< \right]$$

$$= \delta(t-t') + \varepsilon_e G_{re}^<(t, t') + \sum_{r'} t_{re}^* G_{rr'}^<(t, t')$$

In other words, we obtain:

$$\boxed{\varepsilon_e - i}$$

$$\boxed{-i \partial_{t'} G_{re}^<(t, t') = \cancel{\delta(t-t')} + \varepsilon_e G_{re}^<(t, t') + \sum_{r'} t_{re}^* G_{rr'}^<(t, t')} \quad (6)$$

Consider now an equation for the NEGF of the left lead alone:

$H \rightarrow H_L$ only

$$-i \partial_{t'} G_{ee'}^L(t, t') = \delta_{ee'} \delta(t-t') + \theta(t-t') (-i \partial_{t'} G_{ee'}^>) + \theta(t'-t) (-i \partial_{t'} G_{ee'}^<)$$

$$-i \partial_{t'} G_{ee'}^> = +i \langle c_{He}(t) (-i \partial_{t'} c_{He'}^+(t')) \rangle$$

$$= -i \langle c_{He}(t) (+\varepsilon_e c_{He'}^+(t')) \rangle = \varepsilon_e (-i) \langle c_{He}(t) c_{He'}^+(t') \rangle$$

$$= \varepsilon_e G_{ee'}^>(t, t');$$

$$-i \partial_{t'} G_{ee'}^< = +i \langle (-i \partial_{t'} c_{He}^+(t')) c_{He}(t) \rangle = \varepsilon_e (+i) \langle c_{He'}^+(t') c_{He}(t) \rangle$$

$$= \varepsilon_e G_{ee'}^<(t, t')$$

so that we obtain:

$$-i \partial_{t'} G_{ee'}^L = \delta_{ee'} \delta(t-t') + \varepsilon_e G_{ee'}^L \quad \text{or} \quad \boxed{(-i \partial_{t'} - \varepsilon_e) G_{ee'}^L(t, t') = \delta_{ee'} \delta(t-t')} \quad (7)$$

This function is diagonal in e, e' .

$$\boxed{\varepsilon_e - i}$$

$$G_{ee'}^<(t, t') = i \langle c_{He'}^+(t') c_{He}(t) \rangle$$

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$$C_{He}(t) = e^{iH_L t} c_e e^{-iH_L t}, \quad H_L = \sum_{e'} \epsilon_{e'} c_{e'}^\dagger c_{e'}$$

$$\frac{d}{dt} C_{He} = e^{iH_L t} (iH_L c_e - i c_e H_L) e^{-iH_L t} = i e^{iH_L t} [H_L, c_e] e^{-iH_L t},$$

$$[H_L, c_e] = \sum_{e'} \epsilon_{e'} [c_{e'}^\dagger c_{e'}, c_e] = \sum_{e'} \epsilon_{e'} (-c_{e'}^\dagger c_{e'} c_e - (c_{e'}^\dagger c_e c_{e'}))$$

$$= -\epsilon_e c_e$$

vs $\frac{dC_{He}}{dt} = -i\epsilon_e C_{He} \rightarrow C_{He}(t) = C_{He}(0) e^{-i\epsilon_e t}$

$\alpha = \epsilon_e$ as $C_{He}(0) = c_e$ from the definition at $t=0$

$C_{He}(t) = c_e e^{-i\epsilon_e t}$

 $\rightarrow$

$c_{He}^\dagger(t) = c_{He}^\dagger e^{i\epsilon_e t}$

 $\leftarrow$

(8)

$$\hookrightarrow G_{ee'}^<(t, t') = i \langle c_{He}^\dagger c_{He} \rangle e^{i\epsilon_e t} e^{-i\epsilon_e t'} e^{-i\epsilon_e(t-t')}$$

with $\langle c_{e'}^\dagger c_e \rangle = \frac{1}{Z} \text{Tr} [e^{-\beta H_L} c_{e'}^\dagger c_e]$

$$= \frac{1}{Z} \sum_{\lambda} e^{-\beta E_{\lambda}} \langle \lambda | c_{e'}^\dagger c_e | \lambda \rangle$$

$|\lambda\rangle$ - a state of the system having N electrons $\rightarrow \epsilon_e$
 we should replace in fact ϵ_e with $\epsilon_e - \mu$.

$$|\lambda\rangle \equiv |\{n_e\}\rangle = |0 \dots 1 \dots 1 \dots 0 \dots 1 \dots\rangle \text{ etc.}$$

and hence

$$\langle \lambda | c_{e'}^\dagger c_e | \lambda \rangle = \delta_{ee'} \langle \lambda | \hat{n}_e | \lambda \rangle = \delta_{ee'} \langle \lambda | \hat{n}_e | \lambda \rangle$$

$$\langle c_{e'}^\dagger c_e \rangle = \frac{\delta_{ee'}}{Z} \sum_{\lambda} e^{-\beta E_{\lambda}} \langle n_e \rangle_{\lambda} = \delta_{ee'} \langle n_e \rangle$$

where $\langle n_e \rangle = [e^{\beta(\epsilon_e - \mu)} + 1]^{-1} \equiv f(\epsilon_e)$

is the Fermi function.

$$\hookrightarrow G_{ee'}^<(t, t') = i \delta_{ee'} \langle n_e \rangle e^{-i\epsilon_e(t-t')} = i \delta_{ee'} f(\epsilon_e) e^{-i\epsilon_e(t-t')} \quad (9)$$

Similarly,

$$G_{ee'}^>(t, t') = -i \langle c_{He}(t) c_{He}^\dagger(t') \rangle = -i e^{-i\epsilon_e t} e^{i\epsilon_e t'} \underbrace{\langle c_e c_{e'}^\dagger \rangle}_{\delta_{ee} - \delta_{ee'}} =$$

$$= -i e^{-i\varepsilon_e t + i\varepsilon_e t'} (\delta_{ee'} - \delta_{ee'} f(\varepsilon_e)) = \cancel{-i \delta_{ee'} e^{-i(\varepsilon_e - \varepsilon_{e'})t}}$$

$$\boxed{G_{ee'}^>(t, t') = -i (1 - f(\varepsilon_e)) e^{-i\varepsilon_e(t-t')} \delta_{ee'}} \quad (10)$$

We can also calculate

$$G_{ee'}^{(r)}(t, t') = \theta(t-t') (G^> - G^<) = \theta(t-t') [-i(1-f) e^{-i\varepsilon_e(t-t')} \delta_{ee'} - i f e^{-i\varepsilon_e(t-t')} \delta_{ee'}] = \delta_{ee'} \theta(t-t') [-i + i f - i f] e^{-i\varepsilon_e(t-t')}$$

$$\boxed{G_{ee'}^{(r)}(t, t') = -i \delta_{ee'} \theta(t-t') e^{-i\varepsilon_e(t-t')}} \quad (\text{retarded}) \quad (11)$$

Advanced:

$$\boxed{G_{ee'}^{(a)}(t, t') = -\theta(t'-t) (G^> - G^<) = +i \theta(t'-t) \delta_{ee'} e^{-i\varepsilon_e(t-t')}} \quad (\text{advanced}) \quad (12)$$

- Similarly for the right lead, So, we know G^L and G^R for both individual leads, Introduce their inverse.

$$\int_C d\tau \int_{e''} G_{ee'}^L(t, \tau) G_{e'e''}^L(\tau, t') = \delta_{ee'} \delta(t-t')$$

From the EOM for

The solution of

$$(-i\partial_{t_i} - \varepsilon_e) G_{re}(t, t') = \sum_{r, r'} G_{rr'}(t, t') t_{r'e}^* \equiv \Lambda_{re}(t, t') \quad (13)$$

can be written as: [see the ~~proof~~ ^{vacuum diagram} but here, but the proof is given at the end of these notes]

$$G_{re}^L(t, t') = \int_C d\tau \Lambda_{re}(t, \tau) G_{ee}^L(\tau, t') \quad (14)$$

$$\text{Indeed, } (-i\partial_{t_i} - \varepsilon_e) G_{ee}^L(t, t') = \delta(t-t') \quad (t, t' \in C)$$

and hence

$$(-i\partial_{t_i} - \varepsilon_e) G_{re}^L(t, t') = \int_C d\tau \Lambda_{re}(t, \tau) (-i\partial_{t_i} - \varepsilon_e) G_{ee}^L(\tau, t') =$$

$$= \int_0^{\infty} d\tau \Lambda_{re}(t, \tau) \delta(\tau - t') = \Lambda_{re}(t, t') \text{ as required.}$$

Thus, we have the solution:

$$G_{rp}(t, t') = \int_0^{\infty} d\tau \sum_{r'} G_{rr'}(t, \tau) t_{r'e}^* G_{ee}^L(\tau, t')$$

But ~~similarly to~~, $G_{rr'} \rightarrow G_{rr'}^R \equiv \delta_{rr'} G_{rr}^R(t, \tau)$
(although this must be an approximation)

$$G_{re}(t, t') = \int_0^{\infty} d\tau G_{rr}^R(t, \tau) t_{re}^* G_{ee}^L(\tau, t') \quad (15)$$

• Using Langreth theorem: ~~etc~~ $\rightarrow$

$$G_{re}^{<L}(t, t') = \int_{-\infty}^{\infty} d\tau \left[G_{rr}^{(r)R}(t, \tau) t_{re}^* G_{ee}^{<L}(\tau, t') + G_{rr}^{<R}(t, \tau) t_{re}^* G_{ee}^{(a)L}(\tau, t') \right] \quad (16)$$

Use

$$G_{ee}^{<L}(t, t') \rightarrow f(\epsilon_e) G_{ee}^L(t, t')$$

In fact, G^R and G^L do not depend on indices r, e ,

$$= \int_{-\infty}^{\infty} d\tau t_{re}^* \left\{ (-i\theta(t-\tau) e^{-i\epsilon_r(t-\tau)}) (i f(\epsilon_e) e^{-i\epsilon_e(\tau-t')}) + \right. \\ \left. + \cancel{(-i\theta(t-\tau) e^{-i\epsilon_r(t-\tau)})} + (i f(\epsilon_r) e^{-i\epsilon_r(t-\tau)}) (i\theta(t'-\tau) e^{-i\epsilon_e(\tau-t')}) \right\}$$

$$= \int_{-\infty}^{\infty} d\tau t_{re}^* \left\{ f(\epsilon_e) \theta(t-\tau) e^{-i\epsilon_r t} e^{i\epsilon_e t'} e^{+i(\epsilon_r - \epsilon_e)\tau} + \right. \\ \left. + f(\epsilon_r) \theta(t'-\tau) e^{-i\epsilon_r t} e^{i\epsilon_e t'} e^{i(\epsilon_r - \epsilon_e)\tau} \right\}$$

$$= \int_{-\infty}^t d\tau t_{re}^* f(\epsilon_e)$$

The $G^<(t, t')$ is a function of $t-t'$ only. Calculate its FT:

$$G_{re}^<(\omega) = \int dt(t-t') e^{+i\omega(t-t')} G_{re}^<(t, t') = t_{re}^* \int dt \int d(t-t') e^{+i\omega(t-t')}$$

$$\left[\frac{1}{2\pi} \int d\omega_1 e^{-i\omega_1(t-\tau)} G_{rr}^{(r)R}(\omega_1) \cdot \frac{1}{2\pi} \int d\omega_2 e^{-i\omega_2(\tau-t')} G_{ee}^{<L}(\omega_2) + (\text{similar}) \right]$$

$$= t_{re}^* \int \frac{d\omega_1 d\omega_2}{(2\pi)^2} \left[G_{rr}^{(r)R}(\omega_1) G_{ee}^{<L}(\omega_2) \underbrace{\int dt e^{-i(\omega_2-\omega_1)\tau}}_{2\pi \delta(\omega_1-\omega_2)} \times \underbrace{\int d(t-t') e^{+i\omega(t-t')}}_{\substack{\text{Because } \omega_1=\omega_2 \\ 2\pi \delta(\omega_1)}} + (2nd \text{ term}) \right]$$

$$= t_{re}^* \int \frac{d\omega_1}{2\pi} \left[G_{rr}^{(r)R}(\omega_1) G_{ee}^{<L}(\omega_1) \underbrace{\int d(t-t') e^{+i(\omega-\omega_1)(t-t')}}_{2\pi \delta(\omega-\omega_1)} + (2nd \text{ term}) \right]$$

$$= t_{re}^* \left[G_{rr}^{(r)R}(\omega) G_{ee}^{<L}(\omega) + G_{rr}^{<R}(\omega) G_{ee}^{(q)L}(\omega) \right]$$

Now we calculate the current:

$$I_L = \frac{2e}{\hbar} \text{Re} \sum_{er} t_{er} \underbrace{G_{re}^<(t, t)}_{\frac{1}{2\pi} \int d\omega G_{re}^<(\omega) e^{-i\omega(t-t')}} \quad \parallel \quad 1$$

$$= \frac{2e}{\hbar 2\pi} \text{Re} \sum_{er} t_{er} \int d\omega G_{re}^<(\omega)$$

$$= \frac{2e}{2\pi \hbar} \sum_{er} \underbrace{t_{er} t_{re}^*}_{\text{def}} \int d\omega \text{Re} \left[G_{rr}^{(r)R}(\omega) G_{ee}^{<L}(\omega) + G_{rr}^{<R}(\omega) G_{ee}^{(q)L}(\omega) \right] \quad (17)$$

Now we have to calculate the FT of $G^{(r)}$, $G^{(q)}$ and $G^<$:

$$G^<(\omega) = \int d\tau e^{+i\omega\tau} G^<(\tau) = \int d\tau e^{+i\omega\tau} i f(\epsilon) e^{-i\epsilon\tau} = i f(\epsilon) \int d\tau e^{i(\omega-\epsilon)\tau} = 2\pi i f(\epsilon) \delta(\omega-\epsilon) \text{ and is purely imaginary.} \quad (18)$$

So,

$$J_L = \frac{2e}{\hbar} \sum_{er} t_{er} t_{re}^* \operatorname{Re} \left[i \overbrace{f_L^{(r)R}(\varepsilon_e)}^{x+iy} f_L(\varepsilon_e) + i \overbrace{f_R^{(q)L}(\varepsilon_r)}^{x+iy} f_R(\varepsilon_r) \right]$$

$$= \frac{2e}{\hbar} \sum_{er} t_{er} t_{re}^* \operatorname{Im} \left[f_L(\varepsilon_e) G_{rr}^{(r)R}(\varepsilon_e) + f_R(\varepsilon_r) G_{ee}^{(q)L}(\varepsilon_r) \right] \quad (19)$$

Then, calculate the imaginary part of $G^{(r)}$ and $G^{(q)}$:

$$G^{(r)}(\omega) = \int d\tau e^{i\omega\tau} G^{(r)}(\tau, 0) = \int d\tau e^{i\omega\tau} (-i\theta(\tau) e^{-i\varepsilon_r\tau})$$

$$= -i \int_0^\infty d\tau e^{i(\omega-\varepsilon_r)\tau} = -i \int_0^\infty d\tau (c + is) = + \int_0^\infty d\tau (ic + s)$$

$$\operatorname{Im} G^{(r)}(\omega) = - \int_0^\infty d\tau \cos(\omega-\varepsilon_r)\tau = -\frac{1}{2} \int_0^\infty d\tau (e^{i(\omega-\varepsilon_r)\tau} + e^{-i(\omega-\varepsilon_r)\tau})$$

$$= -\frac{1}{2} \int_0^\infty d\tau e^{i(\omega-\varepsilon_r)\tau} - \frac{1}{2} \int_0^\infty d\tau e^{-i(\omega-\varepsilon_r)\tau} = -\frac{1}{2} \int_0^\infty d\tau e^{i(\omega-\varepsilon_r)\tau} - \frac{1}{2} \int_0^{-\infty} d\tau' e^{+i(\omega-\varepsilon_r)\tau'}$$

$$= -\frac{1}{2} \int_0^\infty d\tau + \frac{1}{2} \int_{-\infty}^0 d\tau = -\frac{1}{2} \int_{-\infty}^\infty d\tau e^{i(\omega-\varepsilon_r)\tau} = -\frac{1}{2} 2\pi \delta(\omega-\varepsilon_r) = -\pi \delta(\omega-\varepsilon_r)$$

$$\therefore \boxed{\operatorname{Im} G_{rr}^{(r)R}(\omega) = -\pi \delta(\omega-\varepsilon_r)} \quad (20)$$

$$G^{(q)}(\omega) = \int d\tau e^{i\omega\tau} G^{(q)}(\tau, 0) = \int_{-\infty}^0 d\tau e^{i\omega\tau} i\theta(-\tau) e^{-i\varepsilon_e\tau}$$

$$= i \int_{-\infty}^0 d\tau \frac{e^{i(\omega-\varepsilon_e)\tau}}{c+is} = \int_{-\infty}^0 d\tau (ic - s)$$

$$\operatorname{Im} G^{(q)}(\omega) = \int_{-\infty}^0 d\tau \cos(\omega-\varepsilon_e)\tau = +\pi \delta(\omega-\varepsilon_e)$$

$$\boxed{\operatorname{Im} G_{ee}^{(q)L}(\omega) = \pi \delta(\omega-\varepsilon_e)} \quad (21)$$

Therefore,

$$J_L = -\frac{2e}{\hbar} \sum_{er} t_{er} t_{re}^* \left[f_L(\epsilon_e) (\pi) \delta(\epsilon_e - \epsilon_r) + f_R(\epsilon_r) \pi \delta(\epsilon_r - \epsilon_e) \right] \quad (22)$$

$$J_L = \frac{2e\pi}{\hbar} \sum_{er} t_{er} t_{re}^* \left[f_L(\epsilon_e) - f_R(\epsilon_r) \right] \delta(\epsilon_e - \epsilon_r) \quad \left(\begin{array}{c} ? \\ \pi \end{array} \right)$$

(there should be no π !) contains μ_L contains μ_R

- The quantum numbers l (or r) correspond to $n k \sigma$
(n -band, k - k -vector, σ -spin)

$$t_{er} \equiv \langle n k \sigma_e | t | n_r k_r \sigma_r \rangle = \delta_{\sigma_e \sigma_r} \langle n k \sigma_e | t | n_r k_r \rangle$$

So we have here a double sum over k points:

$$\sum_{er} \rightarrow \sum_{k k'} \sum_{n_L n_R} \rightarrow \int d\epsilon \mathcal{D}_L(\epsilon) \int d\epsilon' \mathcal{D}_R(\epsilon') \sum_{n_L n_R} \dots$$

$\downarrow$ DOS of L $\downarrow$ DOS of R

n_L - all other discrete quantum numbers of L

n_R - the same for R

Then $t_{er} \rightarrow t_{n_e n_r}(\epsilon)$, $t_{re}^* \rightarrow t_{n_r n_e}^*(\epsilon)$

Since the two energies are the same anyway:

$$J_L = \frac{2e\pi}{\hbar} \sum_{n_R n_L} \int d\epsilon \int d\epsilon' \mathcal{D}_L(\epsilon) \mathcal{D}_R(\epsilon') [f_L(\epsilon) - f_R(\epsilon')] \delta(\epsilon - \epsilon') t_{n_L n_R}(\epsilon) t_{n_R n_L}^*(\epsilon)$$

$$J_L = \frac{2e\pi}{\hbar} \int d\epsilon \mathcal{D}_L(\epsilon) \mathcal{D}_R(\epsilon) [f_L(\epsilon) - f_R(\epsilon)] \text{Tr}(t_{n_L}(\epsilon) t_{n_L}^\dagger(\epsilon))$$

(23)

~~# I = \dots~~ $f_L(\epsilon) - f_R(\epsilon) =$

-h-

At small V and $T=0$ this is simplified.

$$f_L(\varepsilon) = [e^{\beta(\varepsilon - \mu + eV)} + 1]^{-1}$$

$$f_R(\varepsilon) = [e^{\beta(\varepsilon - \mu)} + 1]^{-1}$$

, $\mu = \varepsilon_F$ (coincides in both metals before the bias is applied)

$$f_L(\varepsilon) = f(\varepsilon + eV), f_R(\varepsilon) = f(\varepsilon),$$

$$\Delta f_{LR}(\varepsilon) = f_L(\varepsilon) - f_R(\varepsilon) = f(\varepsilon + eV) - f(\varepsilon)$$

$$= f(\varepsilon) + \left(\frac{\partial f}{\partial \varepsilon} \right)_{V=0} eV - f(\varepsilon) + \dots = \left(\frac{\partial f}{\partial \varepsilon} \right)_{V=0} eV$$

where

$$\frac{\partial f}{\partial \varepsilon} = -[e^{\beta(\varepsilon - \mu)} + 1]^{-2} e^{\beta(\varepsilon - \mu)} \beta$$

$$= -\frac{\beta e^{\beta(\varepsilon - \mu)}}{(1 + e^{\beta(\varepsilon - \mu)})^2} \sim -\delta(\varepsilon - \varepsilon_F)$$

so that

$$J_L = \left[\frac{2e^2}{h} \pi \mathcal{D}_L(\varepsilon_F) \mathcal{D}_R(\varepsilon_F) \text{Tr}(t(\varepsilon_F) t^\dagger(\varepsilon_F)) \right] V \quad (24)$$

Now we derive (14) from (13). We introduce the inverse $G^{L\mp}(t, t')^{-1}$ so
 $(-i\partial_{t_1} - \varepsilon_0) G_{ee}^L(t, t') = \delta(t, t') \rightarrow \int dt_2 G_{ee}^{L\mp}(t, t_2)^{-1} G_{ee}^L(t_2, t') = \delta(t, t')$

Then multiply the eq. for G^L by its inverse and integrate:

$$(-i\partial_{t_1} - \varepsilon_0) \delta(t, t_1) = G_{ee}^L(t, t_1)^{-1} \leftarrow \text{an explicit expression for } (G^L)^{-1}.$$

Then rewrite (13) as follows:

$$(-i\partial_{t_1} - \varepsilon_0) \int d\mathbf{z} G_{re}(1, \mathbf{z}) \mathbb{E}(\mathbf{z}, 1') = \Lambda(1, 1')$$

$$\int d\mathbf{z} G_{re}(1, \mathbf{z}) G_{ee}^L(\mathbf{z}, 1')^{-1} = \Lambda(1, 1')$$

Multiply both sides by $G_{ee}^L(1', 2)$ and $\int d\mathbf{z}$ we'll get:

$$G_{re}(1, 2) = \int d\mathbf{z} \Lambda(1, \mathbf{z}) G_{ee}^L(\mathbf{z}, 2) \text{ which is (14).}$$